Radio wiring without the breadboard

3.3 V only. The radio's 3.3V pin goes to the ESP32 pin marked 3V3, never VIN or 5V. Antenna stays off for these tests. USB unplugged while you wire.

What you need

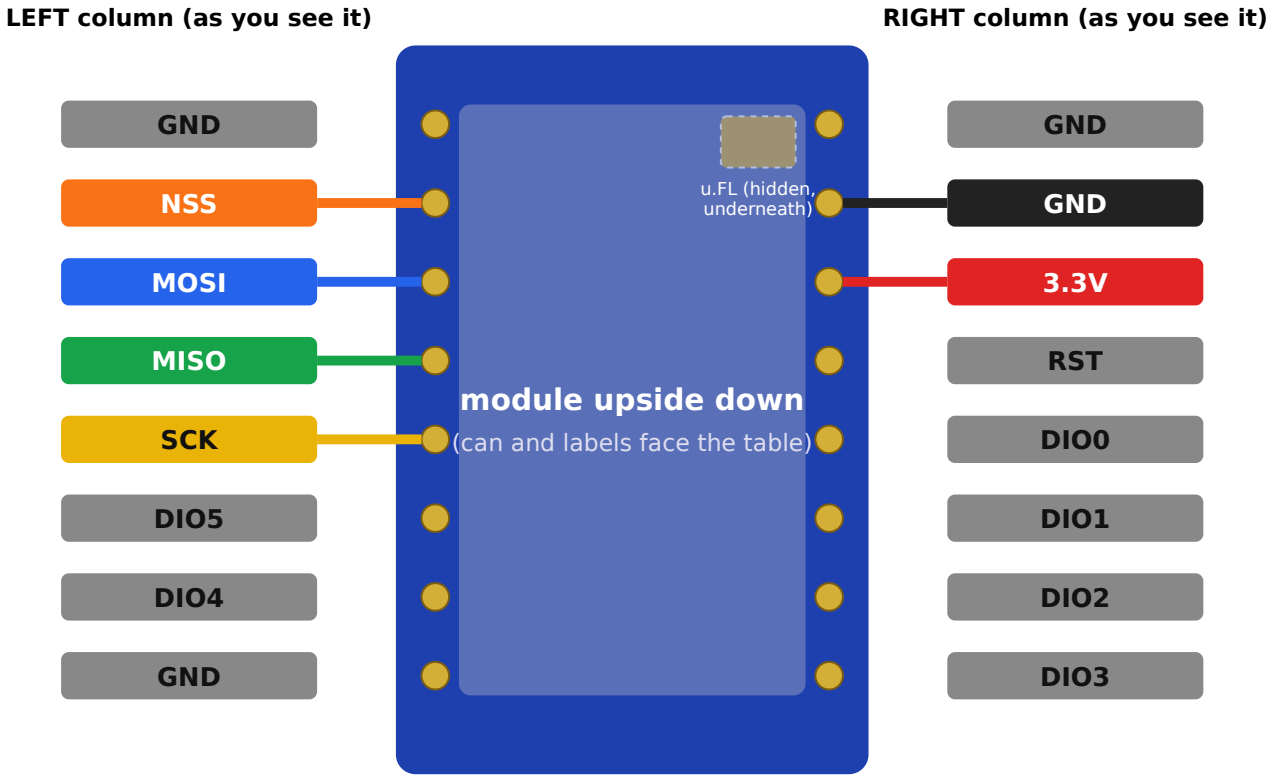
6 male-to-female jumper wires in different colours (red, black, orange, yellow, green, blue), the Ra-02 radio, the ESP32 on its breadboard, and a flat surface (a piece of cardboard works).

1. Take the radio out of the breadboards

Pull it straight out, rocking gently, and remove all the old wires from the radio side. Keep the ESP32 where it is.

2. Flip the radio and lay it flat

Turn it over **like turning a page from left to right**, so the **pins point up** and the metal can faces the table. Rest it on the cardboard. The printed labels are now underneath, so use the picture below instead.



The columns are mirrored compared with the printed labels, because you flipped it. The top-to-bottom order is the same. The thing to remember is that the **3.3V and GND wires go on the column on your right**, and the four signal wires go on the column on your left.

3. Connect the six wires (female end onto the pin)

Wire	Radio pin (female end)	ESP32 pin (male end)
Red	3.3V (right column, 3rd pin)	3V3

Black	GND (right column, 2nd pin)	GND
Orange	NSS (left column, 2nd pin)	D5
Blue	MOSI (left column, 3rd pin)	D23
Green	MISO (left column, 4th pin)	D19
Yellow	SCK (left column, 5th pin)	D18

Push each female end **fully down** onto its pin. Hold the radio with one hand so you don't stress the solder joints. Wires must not touch each other's metal.

4. ESP32 end

- **3V3 and GND:** top two rows of the left row (next to the USB port). Red at 3V3, black at GND.
- **D5, D18, D19:** the stack you built before: orange, yellow, green in **consecutive rows touching each other**. Check the printed labels.
- **D23:** the **last pin at the bottom** of the left row (blue).

5. Test

Plug in USB, press **EN**, and tell me "female wired". I'll run the chip-ID test. The result I'm looking for is **20/20 on every line with 0x12**.

If the grip is loose

Squeeze each female end slightly with pliers so it grips tighter. If a pin still won't hold, we solder that wire on.